



COMPUTER SCIENCE

CS-305L: Parallel & Distributed Computing Lab

SUBMITTED BY

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SECTION : CS-A
SEMESTER : 6th

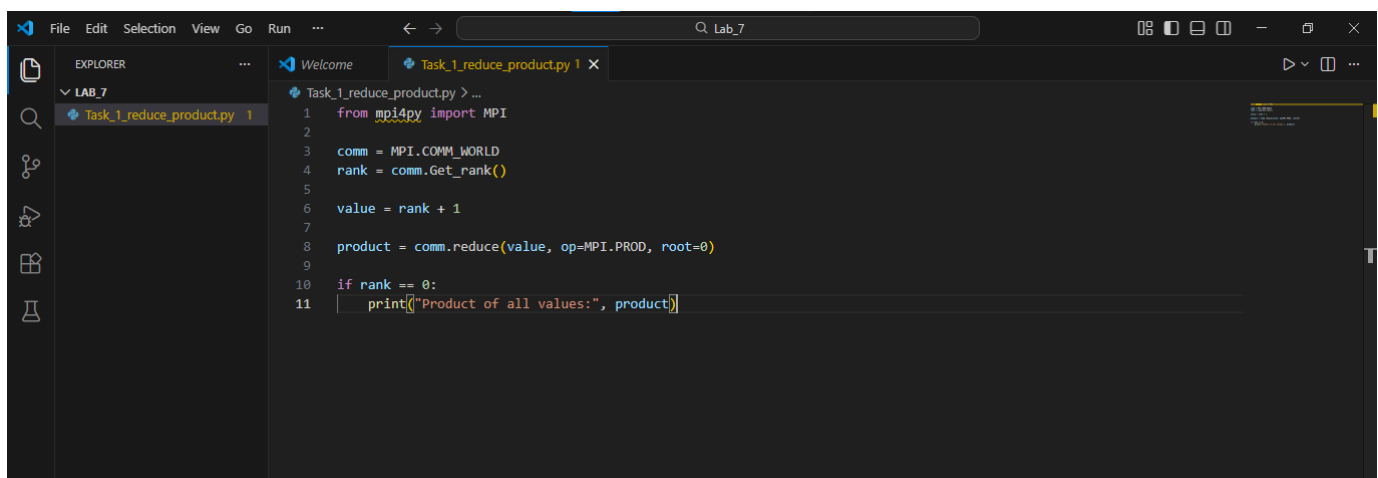
Date: 3/11/2026

Lab 07

Task 1

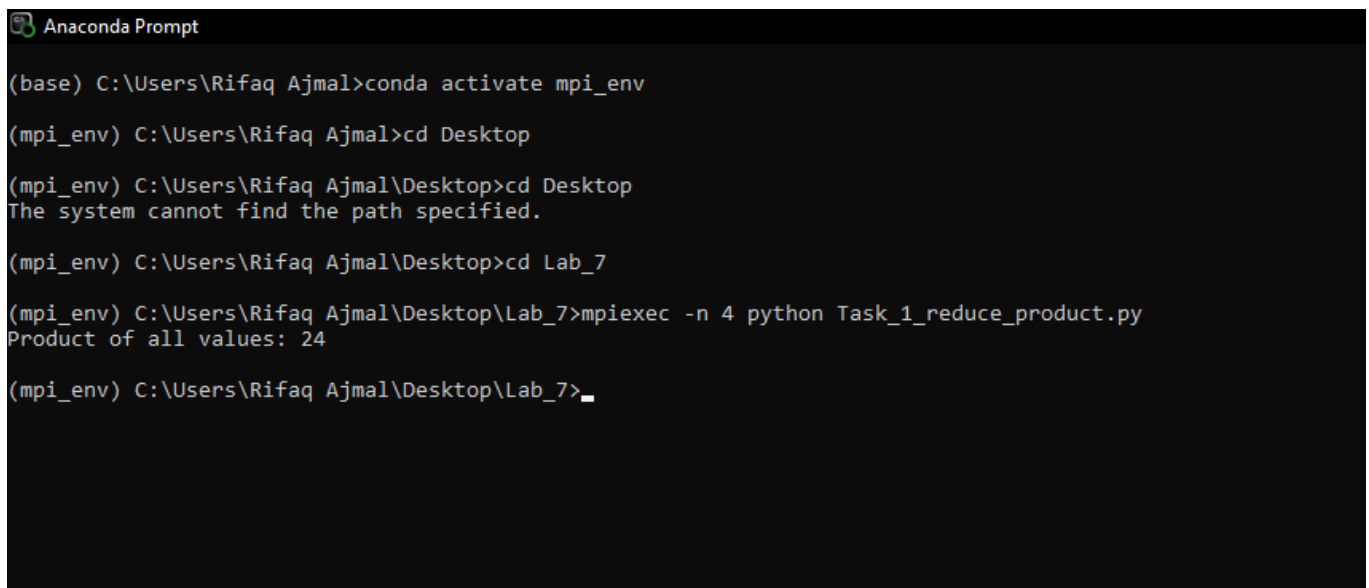
Modify MPI_Reduce to compute product instead of sum

Code



```
1 from mpi4py import MPI
2
3 comm = MPI.COMM_WORLD
4 rank = comm.Get_rank()
5
6 value = rank + 1
7
8 product = comm.reduce(value, op=MPI.PROD, root=0)
9
10 if rank == 0:
11     print("Product of all values:", product)
```

Output



```
Anaconda Prompt

(base) C:\Users\Rifaq Ajmal>conda activate mpi_env

(mpi_env) C:\Users\Rifaq Ajmal>cd Desktop

(mpi_env) C:\Users\Rifaq Ajmal\Desktop>cd Desktop
The system cannot find the path specified.

(mpi_env) C:\Users\Rifaq Ajmal\Desktop>cd Lab_7

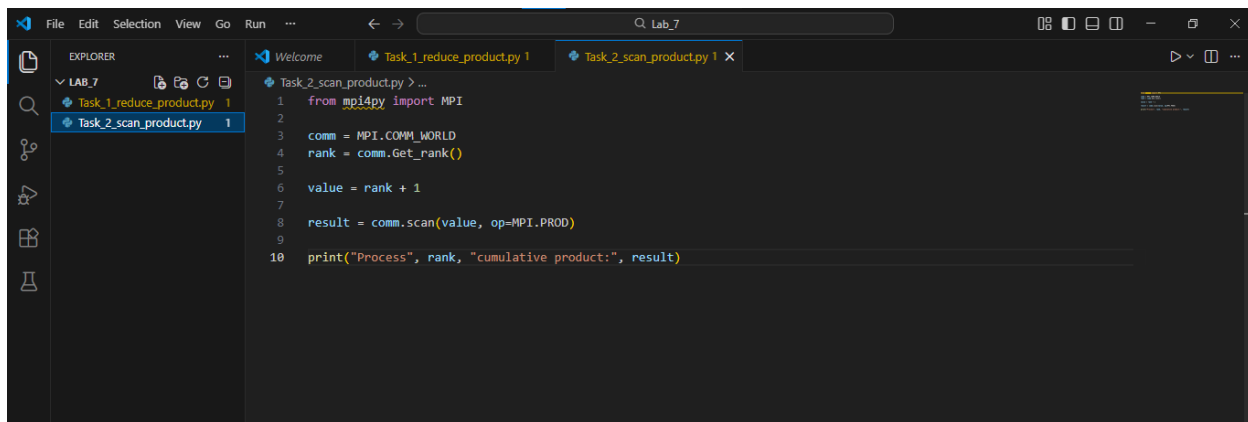
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_7>mpiexec -n 4 python Task_1_reduce_product.py
Product of all values: 24

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_7>_
```

Task 2

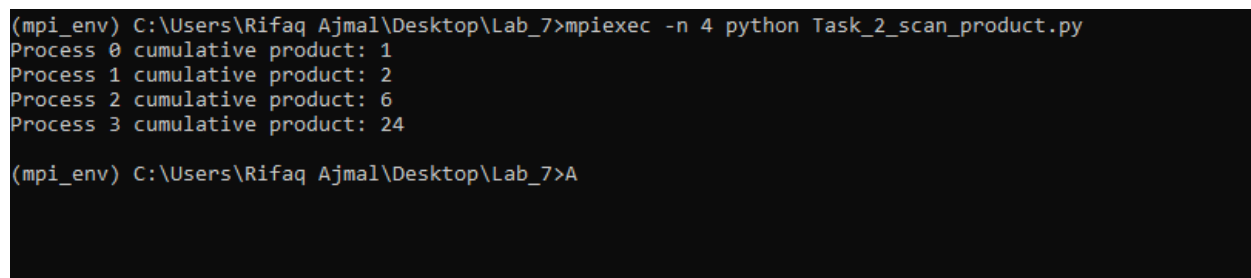
Implement MPI_Scan for cumulative product

Code



```
1 from mpi4py import MPI
2
3 comm = MPI.COMM_WORLD
4 rank = comm.Get_rank()
5
6 value = rank + 1
7
8 result = comm.scan(value, op=MPI.PROD)
9
10 print("Process", rank, "cumulative product:", result)
```

Output



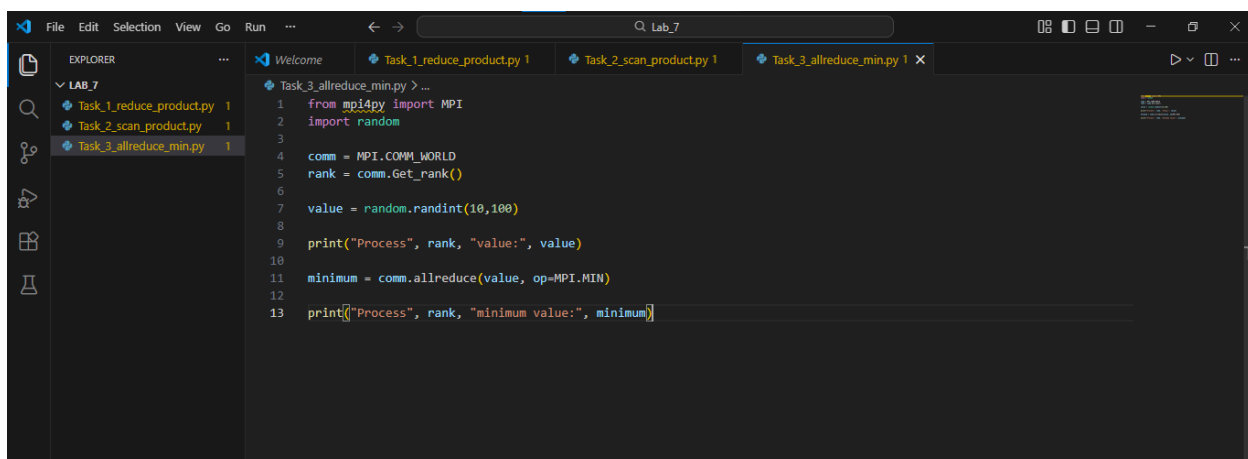
```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_7>mpiexec -n 4 python Task_2_scan_product.py
Process 0 cumulative product: 1
Process 1 cumulative product: 2
Process 2 cumulative product: 6
Process 3 cumulative product: 24

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_7>A
```

Task 3

Program to find minimum value using MPI_Allreduce

Code



```
1 from mpi4py import MPI
2 import random
3
4 comm = MPI.COMM_WORLD
5 rank = comm.Get_rank()
6
7 value = random.randint(10,100)
8
9 print("Process", rank, "value:", value)
10
11 minimum = comm.allreduce(value, op=MPI.MIN)
12
13 print("Process", rank, "minimum value:", minimum)
```

Output

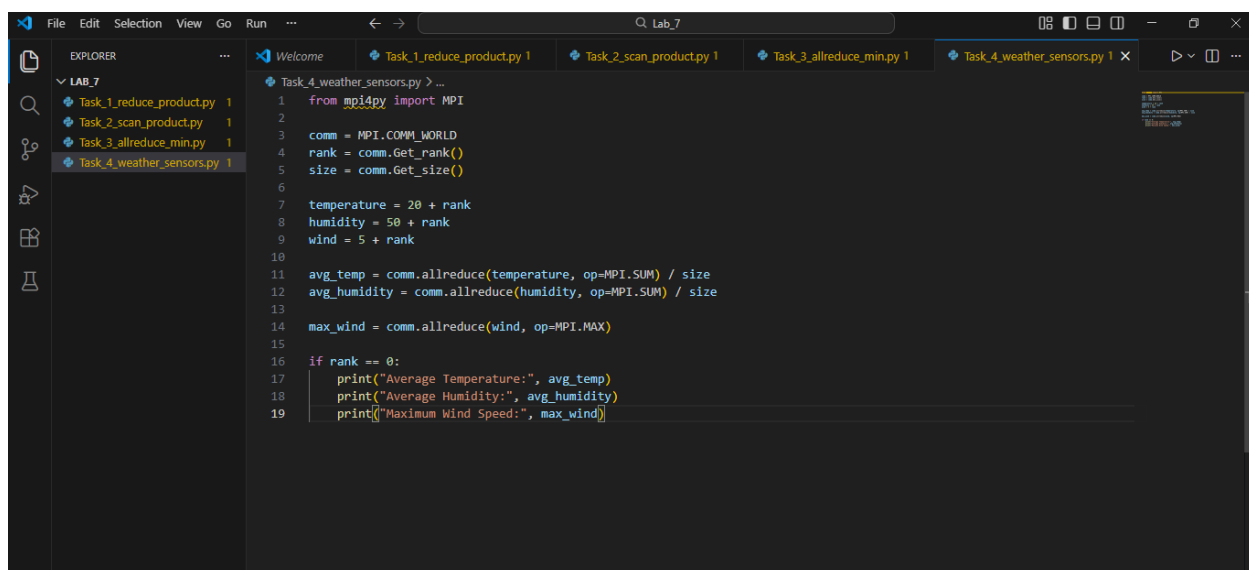
```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_7>mpiexec -n 4 python Task_3_allreduce_min.py
Process 2 value: 81
Process 2 minimum value: 11
Process 1 value: 63
Process 1 minimum value: 11
Process 3 value: 63
Process 3 minimum value: 11
Process 0 value: 11
Process 0 minimum value: 11

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_7>_
```

Task 4

Weather sensor example with humidity and wind speed

Code



```
1 from mpi4py import MPI
2
3 comm = MPI.COMM_WORLD
4 rank = comm.Get_rank()
5 size = comm.Get_size()
6
7 temperature = 20 + rank
8 humidity = 50 + rank
9 wind = 5 + rank
10
11 avg_temp = comm.allreduce(temperature, op=MPI.SUM) / size
12 avg_humidity = comm.allreduce(humidity, op=MPI.SUM) / size
13
14 max_wind = comm.allreduce(wind, op=MPI.MAX)
15
16 if rank == 0:
17     print("Average Temperature:", avg_temp)
18     print("Average Humidity:", avg_humidity)
19     print("Maximum Wind Speed:", max_wind)
```

Output

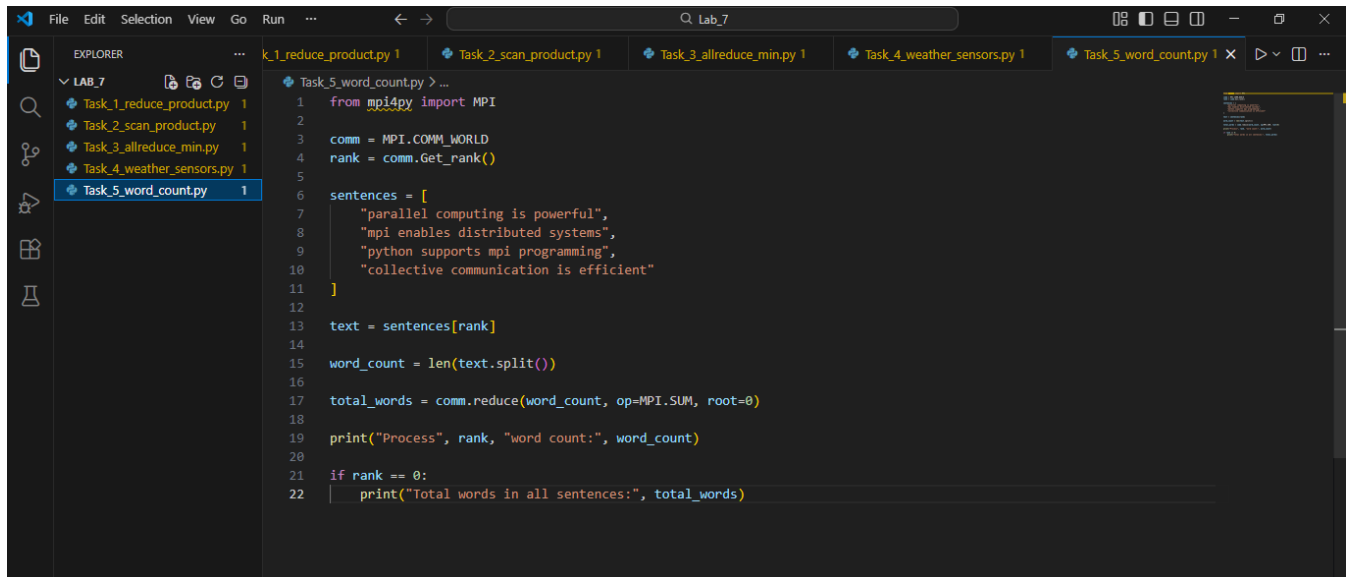
```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_7>mpiexec -n 4 python Task_4_weather_sensors.py
Average Temperature: 21.5
Average Humidity: 51.5
Maximum Wind Speed: 8

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_7>_
```

Task 5

Distributed Word Count using MPI

Code



```
1 from mpi4py import MPI
2
3 comm = MPI.COMM_WORLD
4 rank = comm.Get_rank()
5
6 sentences = [
7     "parallel computing is powerful",
8     "mpi enables distributed systems",
9     "python supports mpi programming",
10    "collective communication is efficient"
11]
12
13 text = sentences[rank]
14
15 word_count = len(text.split())
16
17 total_words = comm.reduce(word_count, op=MPI.SUM, root=0)
18
19 print("Process", rank, "word count:", word_count)
20
21 if rank == 0:
22     print("Total words in all sentences:", total_words)
```

Output

```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_7>mpiexec -n 4 python Task_5_word_count.py
Process 3 word count: 4
Process 2 word count: 4
Process 0 word count: 4
Total words in all sentences: 16
Process 1 word count: 4

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_7>
```